Landscapes, forests, and rivers from Egypt to Central Asia and Kashmir, 3500–300 BCE

Overview

The most useful reconstruction is not "a formerly green Middle East that gradually became desert," but a mosaic of enduring mountain, steppe, desert, forest, and wetland environments whose boundaries and intensity changed locally. Southern Anatolia and much of Arabia experienced long-term drying, whereas some westerly-influenced Central Asian records show increasing effective moisture; human clearance and grazing also altered vegetation independently of climate (Woodbridge et al., 2019, Petraglia et al., 2020, Peng et al., 2024).

Forest cover and river behavior must be reconstructed separately: a treeless desert could contain a substantial mountain-fed river, while a wooded mountain region could drain into a landlocked salt lake or an oasis whose channels shifted frequently. Contrasting examples include the Nile, the Murghab alluvial fan, the Sarazm–Zerafshan valley, and Lake Maharlou in the Zagros (Peeters et al., 2024, Arciero, 2024, Spengler and Willcox, 2013, Saeidi Ghavi Andam et al., 2021).

This report follows a broad geographical circuit: Egypt and Sinai; the Levant and Anatolia; northern Iran and southern Central Asia; Tajikistan, Afghanistan, and Kashmir; the Indus-facing return route and Iranian plateau; Mesopotamia; and the Arabian Peninsula. Modern country names are used only as locators, and the discussion distinguishes directly dated ancient evidence from geographical context and explicitly labeled inference.

How to interpret the reconstruction

Dates, evidence, and confidence

For orientation, approximate conversions used here are: 5,450 calibrated years BP ≈ 3500 BCE; 5,200 ≈ 3250 BCE; 4,200 ≈ 2250 BCE; 3,200 ≈ 1250 BCE; and 2,250 ≈ 300 BCE. "BP" in calibrated radiocarbon chronologies is referenced to 1950, not to the present year; dates reported simply as "years ago," OSL ages, and uncalibrated radiocarbon ages require more care and are retained approximately where necessary (Heinecke et al., 2018, Peeters et al., 2024).

- **Pollen:** Useful for identifying vegetation and regional change, but the percentage of tree pollen is not the percentage of land under forest, and cultivated trees can complicate interpretation of "woodland" signals (Roberts et al., 2019).
- **Wood charcoal:** Identifies trees people obtained and burned, but collection preferences, transport, and preservation mean that charcoal is not a direct census of standing forest (Spengler and Willcox, 2013).
- **Sediments and buried channels:** Can establish former waterways and depositional environments, but their dates and spatial coverage determine whether an exact ancient channel can be reconstructed (Singh et al., 2017, Peeters et al., 2024).

- **Settlement and irrigation evidence:** Demonstrates human use and modification of water, but does not by itself establish regional rainfall, forest extent, or the cause of settlement abandonment (Arciero, 2024, Spate et al., 2022).

Confidence terminology: "Strong" means directly dated local evidence supports the stated change; "moderate" means a regional synthesis or indirect evidence supports it; "limited" means the available material cannot securely reconstruct the whole requested interval. These are assessments for this report, not formal statistical confidence intervals.

River forms that matter

The distinction between different river forms is important because the ancient Nile, an oasis fan, a mountain valley, and a wadi did not create interchangeable landscapes. The following terms describe the forms discussed in the regional studies (Peeters et al., 2024, Giosan and Goodman, 2025, Létolle et al., 2007).

Term	Meaning for reconstructing an ancient landscape
Meandering channel	A winding river whose bends migrate, producing point bars, cutoffs, and abandoned loops.
Braided or wandering river	A mobile channel belt with bars and changing flow paths; not necessarily a permanently fixed single river.
Anabranching/anastomosing system	Several channels divided by floodplain or islands; these terms should not be treated as synonymous with braiding in every study.
Avulsion	A relatively abrupt switch of flow into a different route across a floodplain or fan.
Distributary	A branch carrying water away from the main river toward a delta, lake, or terminal wetland.
Terminal alluvial fan or inland delta	A branching sedimentary system that disperses water inland rather than necessarily reaching the sea.
Endorheic basin	A drainage basin whose water terminates inland, commonly in a lake, wetland, or salt flat.
Wadi	A valley or watercourse commonly associated with episodic runoff; the term alone does not establish its ancient flow duration.
Aggradation / incision	Respectively, sediment buildup raising a river or floodplain surface, and erosion cutting downward into it.

Regional circuit

Egypt: Nile Valley, Delta, and desert margins

Terrain and vegetation. Egypt's principal contrast was between the river-supported floodplain and the surrounding arid landscape, not between a countrywide forest and a later desert. Delta and Memphite reconstructions describe marshes, interconnected waterways, levees, and increasingly well-drained settlement and agricultural surfaces; the Giza record documents a former river branch and floodplain wetlands rather than a closed forest (Bunbury and colleagues, in Willems and Dahms, 2017, Sheisha et al., 2022).

Wooded margins in early Upper Egypt. At Adaïma, occupied from the mid-fourth into the early third millennium BCE, charcoal research reconstructs probably more extensive vegetation and

more wooded desert margins when the settlement was established (Newton and Midant-Reynes, 2007). Changes over about six centuries suggest possible local desertification and human effects on floodplain vegetation, including the spread of disturbance-associated Calotropis procera; this is a local inference, not evidence that all Egypt was densely forested (Newton and Midant-Reynes, 2007).

Delta change near the beginning of the interval. Around 4000–3000 BCE, an earlier network of many interconnecting channels and extensive marsh habitats began giving way to more discrete meandering branches radiating from the delta head, with marshes increasingly concentrated toward the coast. This was a transition, not the disappearance of all wetlands or the establishment of today's two-branch delta (Bunbury and colleagues, 2017).

Luxor: an important correction to the modern picture. Sediment cores reconstruct a dynamic, wandering-braided Nile near Luxor until approximately 4,000 years ago, followed by less-dynamic straight/anabranching channels and rapid floodplain aggradation. The present, largely single-thread local river was established only around 2,000 years ago, after the period covered here; therefore a modern single-channel Nile is a poor template for much of 3500–300 BCE (Peeters et al., 2024).

Why drying could enlarge useful floodplain. At Luxor, declining discharge and increased fine-sediment supply are proposed to have shifted the river from incision toward deposition, expanding the floodplain and its fertile soils. This is a case in which regional drying did not simply mean less arable land, although the study cautions against assigning precise political consequences to dates with substantial uncertainty (Peeters et al., 2024).

Giza: branch-level fluctuations. The reconstructed Khufu branch declined after the local end of the African Humid Period, remained sufficiently active during Old Kingdom pyramid construction, reached a low around 2225 ± 80 BCE, recovered during the Middle Kingdom, and declined again after about 1350 ± 80 BCE. These fluctuations show that “the Nile became drier” conceals substantial differences between centuries and branches (Sheisha et al., 2022).

Later landscape. A reconstruction for Middle Egypt around 1000 BCE identifies meanders, natural levees, point bars, islands, oxbows, low flats, and wet depressions associated with the Nile and Bahr Yusuf. It supports a locally intricate river landscape rather than a uniformly narrow, straight corridor (Antoine, in Willems and Dahms, 2017).

Confidence and gap: Strong for the dated Luxor and Giza examples, moderate for the Delta-wide transition; the fetched evidence does not provide a defensible Egypt-wide forest-cover percentage or a continuous local tree-species history through 300 BCE.

Sinai and the southern Levant: a sharp ecological transition

Terrain and forest pattern. Southern Levantine evidence describes a gradient between Mediterranean woodland/maquis, semi-arid steppe, and desert, with a pronounced rain-shadow contrast across the central ridge. Ancient pollen records document oak, pine, olive, and changing total Mediterranean tree pollen, rather than evidence that the entire southern Levant or Sinai was uniformly forested (Langgut and Finkelstein, “Paleo-environment of the Southern Levant”).

Chronology within 3500–300 BCE. The pollen synthesis identifies Early Bronze I, approximately 3600–3000 BCE, as the wettest Bronze/Iron Age phase in its records, followed by somewhat reduced tree pollen but still relatively humid conditions during much of 3000–2500 BCE. It identifies drying around 2100–1800 BCE, wetter conditions around 1750–1550 BCE, severe drying around 1250 through the late twelfth century BCE, and increased tree pollen around 1100–750 BCE (Langgut and Finkelstein).

Forest versus orchard. Olive pollen is an important indicator of cultivation, so more tree pollen need not mean more natural forest. Conversely, a reduction can reflect orchard abandonment, clearance, grazing, or climatic stress, and these causes must be evaluated rather than assumed (Langgut and Finkelstein, Roberts et al., 2019).

River and lake difference from Egypt. The southern Levantine reconstruction is organized around the Jordan–Galilee–Dead Sea system and smaller catchments, and its dated sedimentary changes often concern lake levels rather than channel relocation. It does not establish an ancient Nile-like floodplain network across the region or a securely dated sequence of Jordan channel positions (Langgut and Finkelstein).

Confidence and gap: Strong for the cited pollen sequences through approximately 600 BCE, but limited for exact river courses, Sinai-specific vegetation histories, and an extension of these records to 300 BCE. A separate Sinai reconstruction would require more local cores and archaeological environmental evidence.

Lebanon, the Beqaa, and the Orontes corridor

Forest composition and spatial variation. Southern Beqaa wetland records contain cedar, pine, and both deciduous and evergreen oak signals, but neighboring catchments did not share an identical clearance history. From approximately 3,500 to 2,000 cal BP, Aammiq suggests reduced deforestation on the adjoining Mount Lebanon slope, while Chamsine/Anjar records continued deciduous-oak expansion and the valley also shows grazing indicators (Hajar et al., 2009).

A necessary qualification about cedars. Timber exploitation was real, but "ancient empires steadily removed all Lebanon's cedar forest" is too simple: one local record can show reduced disturbance while another area remains exploited. The two-core study explicitly leaves room for different histories on other slopes, and it does not justify a continuous region-wide decline throughout 3500–300 BCE (Hajar et al., 2009).

Terrain and water. The Orontes corridor contains mountain-bounded plains, gorges, and marsh basins rather than a single open floodplain; its geography constrained where irrigation and movement were possible. Its spring-fed reaches and the Ghab depression provide a very different setting from the lower Euphrates or Nile Delta, although the geographical descriptions alone do not date ancient channel changes (Turri, "Geopolitics of the Orontes valley").

Bronze Age opening around Qatna. A regional synthesis reports that original oak forest near Mishrife/Qatna had already been cleared by the beginning of Early Bronze IV, followed by secondary shrubland/juniper vegetation in the Middle Bronze Age and a more open landscape

with scattered deciduous oaks around the Middle-to-Late Bronze transition. This is a local trajectory, not a map of all Syria (Turri).

Confidence and gap: Moderate for these local forest trajectories; limited for ancient river alignments and a complete Lebanon–Syria forest map.

Anatolia: wooded uplands, open interior, and changing coasts

The main geographical divisions. Southern Anatolian research distinguishes a coastal zone, the Taurus Mountains with intramontane lake basins, and the inner plateau. Wetter southwestern uplands supported mixed conifer–deciduous woodland, while the interior included oak parkland, wood pasture, and open vegetation; even in antiquity, "Turkey" was not one continuous, uniformly dense forest (Woodbridge et al., 2019).

Long-term change. The southern Anatolian synthesis indicates generally wetter conditions until about 5,000 cal BP, followed by a drier interval encompassing much of the period discussed here. Central Anatolian oak parkland persisted until about 4,000 cal BP in the regional synthesis, after which pasture and cultivated-tree landscapes became more prominent (Woodbridge et al., 2019, Roberts et al., 2019).

Human transformation intensified unevenly. Cultivated-tree indicators increased from about 5,000 cal BP, and the Beyşehir Occupation Phase became a major late-Holocene signature of pasture, cereals, orchards, and declining forest cover, with different onset dates among basins. Its broad development from approximately the later second millennium BCE onward should not be applied as a single synchronous clearance date for all Anatolia (Woodbridge et al., 2019).

Northwestern Anatolia. At Lake Iznik, clear evidence for crops, cultivated trees, disturbance, and forest clearance appears from around 4.8 ka cal BP, approximately the third millennium BCE. The record subsequently alternates between intensive use and vegetation recovery, rather than recording only irreversible loss (Miebach et al., 2016).

Western coastal rivers. The Küçük and Büyük Menderes systems progressively filled marine gulfs with delta and floodplain sediment over the last seven millennia, transforming former ria coasts. Consequently, present shorelines and inland distances are unsafe templates for Bronze and early historical western Anatolia; the timing and position of each shoreline still require site-specific reconstruction (Brückner, 2019).

Confidence and gap: Strong-to-moderate for the sampled southwestern, central, and Iznik basins, but the southern synthesis explicitly underrepresents coastal plains and cannot supply a detailed Pontic/Black Sea forest history. A region-wide reconstruction should leave those gaps visible rather than extend the central plateau pattern northward.

Northern Iran and the southern Caspian: a major forest exception

A distinct wooded belt. The Hyrcanian region is an essential correction to the idea that the route east from Anatolia simply becomes progressively treeless. A central northern Iranian pollen reconstruction indicates that beech–hornbeam forests became widely established after about 8,600 cal BP and persisted afterward, so their existence long predates 3500 BCE (Ramezani Kakroudi, 2024).

Forest types and change. The same reconstruction includes oak, beech, hornbeam, wingnut, and other deciduous trees, with a wingnut increase around 3,400–1,100 cal BP interpreted as a warm/wet late-Holocene interval. Only the earlier portion of that broad interval belongs before 300 BCE; later conditions must not be projected backward (Ramezani Kakroudi, 2024).

Hydrological differences. At the southern Caspian lowland/foothill site, a highstand beginning around 2,600 cal BP affected tree growth through inundation, while the eventual culmination after 2,100 cal BP lies outside this report's endpoint. This was a forest–wetland–lake-margin system, not a Central Asian desert oasis (Ramezani Kakroudi, 2024).

Confidence and gap: Moderate, based here on a doctoral thesis synthesis of several pollen records; exact ancient river channels are not provided. Local human disturbance existed, but the evidence does not justify treating the entire Caspian forest belt as cleared by 300 BCE.

Turkmenistan: foothill streams, desert, and the Murghab oasis

Woodland was concentrated, not continuous. Archaeobotanical work at Anau North describes a Chalcolithic landscape with tugai/tugay vegetation, especially tamarisk, beside a stream, with probable irrigated fields and nearby steppe grazing. Its fifth- and fourth-millennium BCE chronology partly overlaps the beginning of this report, but it is not a vegetation map for all Turkmenistan or the later Iron Age (Miller, University of Pennsylvania research archive).

What tugai means here. It is river-associated woody vegetation within an otherwise much drier landscape, not evidence of extensive rainfall-fed forest across the desert. At Anau, the interpretation places cultivated land, streamside thickets, and steppe in close proximity (Miller).

Murghab river form. The Bronze Age Murghab landscape was an alluvial fan with channels that could remain stable in one area but become unreliable in another. A study of 2400–1300 BCE finds that large settlement clusters around Togolok used relatively stable channels and diverse crops, whereas Ojakly communities faced greater hydrological instability and used mobility and lower-investment farming (Arciero, 2024).

Change through time. The Murghab evidence supports increasingly arid conditions and locally changing water access, but it also documents multiple adaptations rather than a uniform disappearance of the oasis. A settlement could lose its water supply because its particular branch changed even when other parts of the fan remained productive (Arciero, 2024).

Confidence and gap: Strong-to-moderate for the studied Bronze Age localities; limited for a complete fan-wide channel map from 3500 to 300 BCE.

Lower Oxus, Khwarazm, and the Uzboy problem

A branching terminal system, not one immutable river line. The lower Amu Darya region includes the Sarykamysh depression, western channels, the Uzboy, and routes toward the Aral basin; geomorphological work establishes repeated drainage changes over the Holocene. However, the detailed hydrological paper examined here concentrates substantially on later episodes and does not securely date a continuous Uzboy flow sequence to 3500–300 BCE (Létolle et al., 2007).

What can be said safely. Water could be partitioned among branches, depressions, lakes, and desert channels, and not every diversion toward the west necessarily meant the entire Oxus reached the Caspian. The proposition that "the Oxus flowed to the Caspian throughout the Bronze Age" is therefore not established by this evidence (Létolle et al., 2007).

Confidence and gap: Strong for the existence of a complicated and changeable drainage landscape, limited for exact courses in particular centuries of the requested period. Any ancient map should use dated alternatives or uncertainty bands, not a single confident Caspian-bound line.

Tajikistan: productive valleys versus the high Pamir

Sarazm and the Zerafshan valley. Sarazm occupied an elevated alluvial terrace in a foothill setting, with agricultural land and access to mountain environments. Fourth- and third-millennium BCE charcoal includes willow/poplar, ash, Russian olive, tamarisk, hackberry, wild almond, pistachio, and juniper, alongside reeds, indicating access to several woody and wetland habitats rather than a uniformly treeless landscape (Spengler and Willcox, 2013).

Likely ancient-to-modern difference. The authors infer that some woodland taxa probably grew closer to the settlement than they do today and discuss subsequent lower-valley deforestation, potentially involving field clearance, fuel and construction demand, and livestock inhibition of regeneration. The rate and timing of that loss are not sufficiently resolved to place all of it securely before 300 BCE (Spengler and Willcox, 2013).

The eastern Pamir was fundamentally different. Lake Karakul lies in a high, arid plateau setting, and its pollen record indicates predominantly open vegetation rather than extensive forest. From about 5.4 to 1.0 cal kyr BP, increased wormwood relative to desert indicators suggests steppe and somewhat greater moisture than during the early Holocene; most of 3500–300 BCE falls within this broad phase (Heinecke et al., 2018).

Water without forest. Karakul's catchment is supplied by seasonal snow/glacier-fed streams in the modern geographical description, and the study suggests changing riverine wetland representation without reconstructing ancient channel positions. Crucially, a meltwater-fed lake can change differently from surrounding dry-land vegetation, so lake level cannot simply be translated into regional rainfall or forest extent (Heinecke et al., 2018).

Confidence and gap: Strong for identified Sarazm wood taxa and the Karakul pollen sequence, but moderate for the spatial extent of ancient woodland and limited for exact ancient river courses.

Afghanistan and the Hindu Kush: an important transition, but uneven evidence

Geographical contrast. Modern botanical geography distinguishes much more wooded monsoon-influenced eastern and southeastern mountains from drier northern and western environments, with oak and conifer belts in the east and open juniper formations on northern slopes. This is valuable environmental context, but it is not a directly dated forest map for 3500–300 BCE (de Planhol, Encyclopaedia Iranica).

Limits of ancient inference. Historical statements about abundant Bactrian pistachio in the Hellenistic period suggest substantial woody resources in parts of the north, but the broad dating does not establish their precise distribution at 300 BCE, much less in 3500 BCE. Assertions about a wooded Kabul basin "2,000 years ago" are later than the requested endpoint and cannot be used to reconstruct the Bronze Age (de Planhol).

Drainage connection. Southwestern Hindu Kush runoff was important to the terminal Hamun/Sistan system, tying highland hydrology to lowland wetland and settlement conditions far away. The available Hamun reconstruction supports changing lake extent, but not a complete ancient channel map across Afghanistan (Hamzeh et al., 2016).

Confidence and gap: Limited for an Afghanistan-wide dated vegetation history; moderate for the highland–terminal-basin hydrological connection. Modern altitudinal belts should be labeled contextual, not presented as measured ancient boundaries.

Kashmir: an alluvial basin below forested and grazed mountain slopes

Terrain. The Kashmir Valley combines an alluvial floor, Jhelum floodplain, raised Karewa surfaces, foothills, and higher mountains; Neolithic sites often occupied elevated Karewa locations above the floodplain. Its ancient lake-basin origins are much older than 3500 BCE, so it should not be depicted as a single enormous undrained lake during the period of early farming (Spate, "Reconsidering Archaeological and Environmental Proxies").

Forest and upland vegetation. Middle- and high-altitude records on the western valley flank contain coniferous forest and subalpine tree/shrub signals including birch, willow, juniper, alder, and hazel. Their importance is the contrast between lower forest settings and more open high pastures, not a single uniform Kashmir forest (Spate et al., 2022).

Changes before 300 BCE. At Tosa Maidan, approximately 3700–3000 BP, moderate disturbance, charcoal, and vegetation changes are interpreted cautiously as low-intensity summer herding and burning; another disturbance phase around 2700–2500 BP indicates greater grazing and clearing. At Pari Has, a possible canopy-opening transition around 2300–2100 BP only begins near the endpoint of this report and largely continues afterward (Spate et al., 2022).

Hydroclimate and river limits. Regional records generally suggest weakening precipitation influences between about 4000 and 2000 BP, but local conditions and human land use varied. The studies support an inhabited floodplain–terrace–mountain landscape, not a securely mapped sequence of Jhelum channel migrations during 3500–300 BCE (Spate et al., 2022, Spate, landscape synthesis).

Confidence and gap: Strong-to-moderate for the sampled upland disturbance phases; limited for early-period forest extent across the entire basin and exact river positions.

Indus-facing return route: distinguish active rivers from abandoned valleys

Why include it. The Indus-facing plains are an important comparison when moving south and west from Kashmir: geomorphological research reconstructs large sediment-built river surfaces and a monsoon-sensitive flood regime, unlike the terminal oasis fans of Turkmenistan. A 2012 synthesis proposes that weakening monsoon-driven flooding first improved agricultural stability and later reduced the dependability of inundation farming (Woods Hole Oceanographic Institution, 2012).

The critical Ghaggar–Hakra correction. A later, directly dated study identifies the Ghaggar–Hakra palaeovalley as an abandoned Sutlej course, with the major Himalayan river's diversion completed shortly after about 8 ka BP, well before 3500 BCE. During Indus urban occupation, the valley hosted much lower-energy monsoon-fed streams and floodplain/ponded-water deposition rather than the large Himalayan river that originally cut it (Singh et al., 2017).

Implication. Settlement beside a broad ancient river valley is not proof that a large river still occupied it, and urban decline cannot automatically be blamed on a major river diversion occurring at that time. The 2017 study explicitly cautions that its later sediment chronology cannot resolve a simple causal link between river change and urban decline (Singh et al., 2017).

Confidence and gap: Strong for the local Sutlej palaeochannel chronology, but this research does not provide forest cover for the Indus basin, Makran, or Balochistan.

Iranian plateau, Zagros, and Sistan: several landscapes within one region

Zagros woodland versus interior basins. Lake Maharlou in southwestern Iran provides evidence for oak, pistachio, and almond woody vegetation in an intermountain basin associated with a seasonal saline lake, rather than a humid closed forest everywhere. Its core begins only around the early second millennium BCE, so it cannot describe the entire interval from 3500 BCE (Saeidi Ghavi Andam et al., 2021).

Woodland pressure and cultivated landscapes. Around 2800–1600 cal BP, natural woody taxa reached low values, with timber use, grazing, and climate all considered possible contributors. Within the Achaemenid interval, approximately 550–330 BCE, cultivated-tree diversity increased, including plane and grape indicators and interpreted cypress/pomegranate cultivation; this is evidence for a managed landscape, not necessarily natural reforestation (Saeidi Ghavi Andam et al., 2021).

Water and vegetation proxies can disagree. Maharlou sediment proxies indicate substantial flood activity and positive water balance during much of approximately 3800–2000 cal BP, while vegetation indicators suggest reduced moisture or altered marsh/grassland conditions over an overlapping interval. Such differences can reflect catchment, land-use, and proxy effects rather than an error that can be resolved by choosing whichever signal fits a simple drying narrative (Saeidi Ghavi Andam et al., 2021).

Sistan/Hamun. This was a terminal wetland/lake system sustained by inflowing rivers rather than a coastal delta or forested valley. A reconstruction suggests substantial lake extent during approximately 6–3.8 kyr BP and links Bronze Age settlement with the resources of the lake and

its margins, but its sparse dating makes precise lake-level and climatic-event correlations tentative (Hamzeh et al., 2016).

Confidence and gap: Moderate for the sampled Zagros and Sistan sequences, limited for exact channel histories and a complete Iranian-interior vegetation map. There is no evidential basis here for filling the whole plateau with forest or for assuming that every terminal lake contracted on the same date.

Mesopotamia: northern woodland-steppe and southern river–marsh mosaics

Northern and middle Euphrates environments. Bronze Age charcoal from northern Syria and southern Turkey documents riverine woodland and pistachio woodland-steppe, with human disturbance first affecting river-associated woods and later expanding into woodland-steppe. Near Emar, the latter probably disappeared by the Late Bronze Age, though the study cannot separate climatic drying from clearance decisively (Deckers and Pessin, 2010).

Southern lowland terrain. Lower Mesopotamia was a low-gradient alluvial and coastal landscape with levees, splays, meander belts, distributary channels, marshes, and changing estuarine/deltaic environments. It should not be reconstructed simply as today's dry archaeological plain crossed by two fixed river lines (Giosan and Goodman, 2025).

River-course instability. Tigris and Euphrates avulsions built overlapping sedimentary ridges and changed local access to water. A site could become separated from a major channel through channel relocation, while flood and irrigation pathways also transformed the surrounding land (Giosan and Goodman, 2025).

Coastline and wetlands. The 2025 reconstruction proposes interacting Tigris–Euphrates and Khuzestan delta systems and a progressively restricted marine embayment; it emphasizes uncertainty about the precise proximity of early cities to the sea. It is safer to depict a shifting delta–marsh–estuarine transition than a sharply drawn, universally agreed Sumerian coastline (Giosan and Goodman, 2025).

Human modification and interpretation. Irrigation increasingly made this a managed landscape, but the article's proposal that early coastal Sumer benefited from tidal irrigation remains a hypothesis, not a demonstrated explanation for every city. Its date-grove references concern agricultural vegetation, not evidence of extensive natural forest across the southern plain (Giosan and Goodman, 2025).

Confidence and gap: Strong for basic alluvial processes, moderate for the proposed detailed coastal evolution, and limited for exact river positions at every archaeological date.

Northern, central, and eastern Arabia: desert with unevenly persistent water

Conditions by 3500 BCE. In southeastern Arabia, the return toward aridity began before the requested interval, after roughly 6 ka, and the reduction in inland evidence during the so-called Dark Millennium overlaps its beginning. Northern Arabian basins nevertheless retained or attracted occupation, so one cannot treat every inland area as simultaneously abandoned (Petraglia et al., 2020).

Water and vegetation. The regional review describes dune-bounded basins, playas, wetlands, shallow aquifers, and oases; it documents grassland retreat and desert-shrub expansion in some earlier records, not a peninsula-wide former forest. At Tayma, wetland freshwater persisted until around 5 ka, while cultivation and water management developed amid increasingly arid conditions (Petraglia et al., 2020).

Bronze Age oasis landscapes. Walled oases appeared at Tayma and Qurayyah in the first half of the third millennium BCE, while Khaybar's enclosure is dated approximately 2250–1950 BCE. These systems enclosed settlements, water, crops, and livestock, demonstrating maintained productive enclaves in desert environments rather than proof of extensive natural woodland (Charloux and Alonazi, 2025).

Drying around the late third millennium. Awafi and Wahalah in southeastern Arabia show reduced humidity around 4.1–3.9 ka, associated with lake drying, dune reactivation, and vegetation loss. The review also notes continued occupation at Tayma through the broader 4.2 ka interval, demonstrating that environmental stress did not produce a uniform regional settlement outcome (Petraglia et al., 2020).

Confidence and gap: Moderate-to-strong for particular lake and oasis records, limited for a continuous central-Arabian vegetation history or exact ancient wadi flow durations.

Oman and southeastern Arabia: wadis, aquifers, and increasingly engineered oases

Different hydrology from Mesopotamia. Research at Salut concerns a wadi alluvial plain and adjoining gravelly fans, where agriculture initially used surface irrigation and later underground falaj water collection/distribution. This is not a landscape organized around two large perennial lowland rivers (Cremaschi et al., 2018).

Bronze-to-Iron Age transition. Buried soils document Bronze Age cultivation and soil management, while local falaj evidence securely places underground irrigation in use by the later first millennium BCE. Because the paper gives slightly different formulations for the earliest dates, the safest statement is that such systems are locally demonstrated before 300 BCE, not that every Omani oasis acquired them simultaneously (Cremaschi et al., 2018).

Forest-cover caution. Modern acacia-type savanna, mountain trees, and palm gardens are useful ecological context, but the Salut study does not identify a dated ancient forest composition (Cremaschi et al., 2018). At Al-Khashbah, late-third-millennium BCE deposits contain charcoal identified as Moringa, Ziziphus or fabaceous wood, with Vachellia/Prosopis among the possible identifications; these establish woody-resource use but not a local closed canopy, and the investigators explicitly find preservation inadequate for a robust local vegetation reconstruction (Schmidt et al., 2025).

Confidence and gap: Strong for local cultivated soils and water-management evidence, limited for ancient forest extent and a complete peninsula-wide falaj chronology.

Yemen and southern Arabia: wetter highlands, dry lowlands, seasonal runoff

Relief mattered. Southern Arabian palaeoenvironmental research distinguishes inland desert basins, piedmont valleys, coastal lowlands, and highlands, with lowland water often supplied by runoff from higher terrain. Highland lakes and seasonal river flow persisted longer than some lowland palaeolakes, so the lowlands and uplands cannot be assigned one drying date (Lézine et al., 2010).

Changes overlapping the requested interval. A Saada highland lake phase is dated approximately 5263–4838 cal BP, while the youngest part of the Awafi lake sequence, ending around 5298 cal BP, records increasing aridity. These examples demonstrate different local water histories near the late fourth and third millennia BCE, not a single synchronous loss of all southern Arabian surface water (Lézine et al., 2010).

Cultivation under aridity. From about 5000 cal BP, irrigated piedmont/oasis agriculture became important in the wider southern and southeastern Arabian record, with date-palm gardens, cereals, and channels at some sites. In Yemen, settlement and irrigation evidence show that dry lowlands could remain inhabited through managed seasonal water rather than dependence on permanent large rivers (Lézine et al., 2010).

Tree-type caution. The same paper describes modern lowland acacia–commiphora and upland olive/juniper associations, but these cannot be transferred unchanged onto a 3500 BCE map. Dated ancient forest cover is less securely established than the contrast between upland runoff, lowland aridity, and irrigated settlement (Lézine et al., 2010).

Confidence and gap: Moderate for the hydrological contrasts and dated lake episodes, limited for exact forest boundaries, coastal mangrove extent, and a continuous sequence through 300 BCE.

Comparative timeline

These periods organize the evidence rather than imply simultaneous change across the entire study area. Where the table gives only a regional example, it should not be extended to neighboring countries without local support.

Approximate interval	Western and southern sectors	Central Asian and mountain sectors	Main interpretation
3500–3000 BCE	Delta/Memphite channel and marsh reorganization; relatively wet Early Bronze I southern Levant; Arabian drying already under way (Nile volume, Levant synthesis, Arabian review).	Sarazm's woody-resource record begins within this broad early settlement horizon; Karakul's steppe-favoring phase begins around 5.4 cal kyr BP (Sarazm, Karakul).	Already a mosaic, not an everywhere-green starting point.
3000–2300 BCE	Increasing Anatolian cultivation/clearance; active Giza branch during pyramid construction; oasis expansion in northwestern Arabia (Iznik, Giza, walled oases).	Diverse wooded and cultivated valley settings at Sarazm; oasis agriculture develops within landscapes that were not region-wide forests (Sarazm, Murghab).	More human-managed vegetation and water, but locally different trajectories.

Approximate interval	Western and southern sectors	Central Asian and mountain sectors	Main interpretation
2300–1800 BCE	Giza low water followed by recovery; Levantine dry phase approximately 2100–1800 BCE; southeastern Arabian lake/dune changes; Luxor regime transition around 4 ka (Giza, Levant, Arabia, Luxor).	Murghab's local channel stability varies; broad Central Asian moisture behavior does not match one universal drought curve (Murghab, Central Asian synthesis).	"4.2 ka drought" is a useful research topic, not a complete explanation.
1800–1200 BCE	Wetter phases followed by late-thirteenth/twelfth-century drying in the southern Levant; stronger anthropogenic landscapes in parts of Anatolia; woodland-steppe loss near Emar by the Late Bronze Age (Levant, Anatolia, Euphrates charcoal).	Murghab communities continue different water-access adaptations; early upland pastoral disturbance appears in Kashmir (Murghab, Kashmir).	Climate, grazing, cultivation, and channel instability interact.
1200–300 BCE	Woodland recovery in some Levantine records, continued managed Anatolian landscapes, complex Egyptian floodplains, and Achaemenid cultivated-tree diversity in Fars (Levant, Anatolia, Nile volume, Maharlou).	Kashmir grazing/clearing intensifies at some sites; the high eastern Pamir remains an open-vegetation landscape; Caspian forests remain a distinct wooded zone (Kashmir, Karakul, Hyrcanian thesis).	No defensible single "final deforestation date" for the whole region.

What changed most, and what did not

Forest-cover differences

- **Humid forest belts:** The southern Caspian remained a fundamentally different ecological province from nearby arid interior basins; its ancient beech–hornbeam history cannot be explained as a temporary Bronze Age greening of desert (Ramezani Kakroudi, 2024).
- **Woodland and parkland zones:** Anatolia, the Levant, and southwestern Iran contained mixtures of woods, scrub, fields, pasture, and cultivated trees whose proportions changed through climate and land use (Woodbridge et al., 2019, Langgut and Finkelstein, Saeidi Ghavi Andam et al., 2021).
- **Riverine woody corridors:** Anau and Sarazm show how locally rich woody vegetation could exist beside streams and irrigated land within broader steppe or dry foothill environments (Miller, Spengler and Willcox, 2013).
- **Naturally open highlands:** The high eastern Pamir pollen record is predominantly open; an absence of forest there need not imply human deforestation (Heinecke et al., 2018).
- **Desert oases and gardens:** Arabian oasis cultivation created productive tree-and-crop patches, but a date-palm garden is not evidence that surrounding desert had once supported closed natural forest (Petraglia et al., 2020, Charloux and Alonazi, 2025).

River-course differences

- **Nile:** Both changing branch discharge and local shifts from incision to aggradation mattered; the modern single-channel planform should not be projected uniformly backward (Sheisha et al., 2022, Peeters et al., 2024).

- **Mesopotamian rivers:** Avulsion, levee building, distributary reorganization, and changing marsh/coast relations were central; rainfall at a city was not the sole control on its water supply (Giosan and Goodman, 2025).
- **Murghab and lower Oxus:** Inland branching, terminal water bodies, and uneven local channel stability require fan- or basin-scale reconstruction, with particularly large chronological uncertainty for the ancient Uzboy (Arciero, 2024, Létolle et al., 2007).
- **Western Anatolian rivers:** Delta growth changed the relation between land, ports, and the sea, making today's coastline an unreliable guide to ancient travel and settlement (Brückner, 2019).
- **Indus-related palaeovalleys:** A large abandoned river valley could remain attractive after the large river left; landform size and settlement density do not establish contemporaneous discharge (Singh et al., 2017).
- **Arabian wadis and aquifers:** Episodic runoff, groundwater, wells, and irrigation could support settlements without a permanent major river, and increased engineering need not mean increased rainfall (Petraglia et al., 2020, Cremaschi et al., 2018).

Major uncertainties and claims to avoid

A single region-wide drying curve is not supported. Westerly-influenced Central Asia includes late-Holocene wetting tendencies that differ from the declining summer-monsoon influence documented farther south and east, with substantial disagreement even among Central Asian local records (Peng et al., 2024, Spate et al., 2022).

The 4.2 ka event is not a universal master explanation. Some local records document marked change near the late third millennium BCE, but timing, severity, and societal responses differ; a 2026 conference abstract still frames the consistency of a southwest Asian speleothem drought signal as an open research question rather than a settled finding. That abstract is included only as evidence that the debate remains active, not as a new completed reconstruction (Petraglia et al., 2020, Paine et al., EGU 2026 abstract).

Exact forest percentages are not available from these records. Pollen and charcoal identify types and trajectories but do not justify statements such as “Anatolia was 80% forested in 3000 BCE” without an explicit quantitative land-cover model (Roberts et al., 2019, Spengler and Willcox, 2013).

Major geographical gaps remain. The evidence gathered is particularly thin for a continuous Sinai sequence, the full Pontic/Black Sea belt, Afghanistan-wide ancient forest boundaries, Balochistan/Makran vegetation, central Arabian interiors, and exact Oxus–Uzboy courses. These should be treated as unresolved research areas, not filled with modern vegetation maps or extrapolated from a distant core.

Access and source limitations. A 2005 Upper Egyptian charcoal paper and an eastern Anatolian archaeobotany chapter could not be fetched, so their detailed findings were not used to construct ancient species maps; the separate, accessible 2007 Adaïma study is used above. Several other sources provide only abstracts or institutional summaries, and their spatial and

chronological limits are identified in the relevant sections rather than hidden behind more precise-sounding claims.

Synthesis

The broad circuit crossed several distinct ecological worlds: Egypt's evolving floodplain and delta; Mediterranean woodland–steppe gradients; Anatolian upland woods and increasingly managed interiors; the exceptional humid Caspian forest belt; Central Asian desert oases and wooded valley patches; open high Pamir landscapes; Kashmir's floodplain–forest–pasture mosaic; and the irrigated river, marsh, and wadi systems of the return route (Nile studies, Anatolian synthesis, Hyrcanian reconstruction, Sarazm, Karakul, Kashmir, Arabian review).

Across 3500–300 BCE, the strongest common development was not uniform desertification but the interaction of shifting water availability with expanding cultivation, grazing, timber demand, and water engineering. For historical reconstruction, the safest approach is therefore a sequence of regional landscapes with dated local waterways and explicit uncertainty, rather than one green-to-brown map for the entire 3,200-year span (Roberts et al., 2019, Arciero, 2024, Spate et al., 2022).